

Retina

Data Ontology Layer that reads and understands your business data.

Max and a Dream

retina-silk.vercel.app



PROBLEM STATEMENT

Business operations' friction remains in manual work



Finding data is hard

Staff search the inbox email by email to find one shipment's documents.



Manual processes

Every Bill of Lading is checked by hand, one by one. One miss costs amendment fees or demurrage.



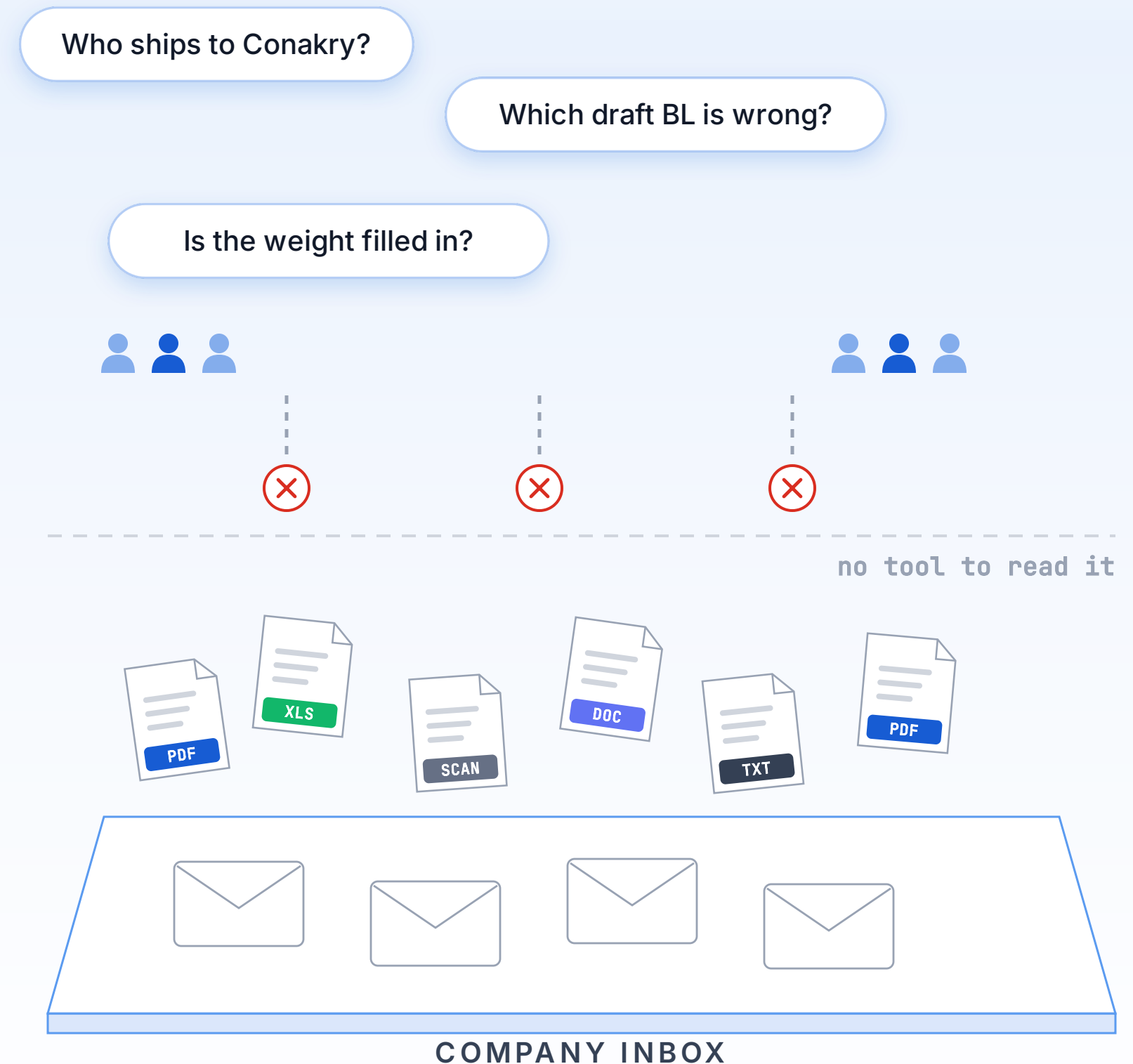
Unstructured, inconsistent data

Values arrive missing or not matching, in different formats and wording.

THE BIGGER PROBLEM

Data already sits in inbox & information, No tool to process and understand it

Companies want zero friction managing their business data. The friction is that most of it never leaves the inbox.



THE PRODUCT

What Retina does



01 Observe

Reads every email and attachment, and keeps the original.



02 Understand

Works out what each email is asking for.



03 Structure

Turns documents into fields, each linked to its source line.



04 Verify

Compares documents and flags what differs or is missing.



05 Answer

Builds a map of companies, ports and shipments anyone can ask.

RE_SI - SIN706562729 - DIRECT(PIL) - 5RCY-72046 - M...

Shipping instruction

The check 0

Model calls 1

so sokyong ooi
sokyong_ooi@aprilasia.com

To ops@aprilasia.com

Hi Lee

Please find Shipping instruction for 5RCY-72046.

POL: RUGAO/NANTONG/SHANGHAI, CHINA
POD: MERSIN, TURKEY

Shipper:
ASIA PACIFIC PAPERBOARD TRADING PTE LTD
80 RAFFLES PLACE, #50-01 UOB PLAZA 1
SINGAPORE 048624

Consignee:
KPP-ANTALIS (SINGAPORE) PTE. LTD.
8 TEMASEK BOULEVARD
#42-01 SUNTEC TOWER 3
SINGAPORE 038988

Notify Party:
KPP-ANTALIS (SINGAPORE) PTE. LTD.
8 TEMASEK BOULEVARD
#42-01 SUNTEC TOWER 3
SINGAPORE 038988

Description of Goods:
15X20'GP
PAPERBOARD

Draft reply

Ask Retina

Write the reply email for email_019, a shippi...

⌕ + ↗

Write the reply email for email_019, a shipping instruction. Put the message in the draft, from the shipment record: what is on file, and what is still blank. Do not recommend an action.

Looking with get_email, run_recipe 16 s

get_email The from: address is the only one the ... 11 ms

run_recipe One row per fact, showing what this ... 18 ms

with members as (
select es.email_id,
es.oc_no, es.bl_no, es.booking_ref, es.inv
es.voyage, es.hs_code, es.container_type,
es.container_count::text as container_coun
es.gross_weight_kg::text as gross_weight_k
... 92 more lines

Reading RE_SI - SIN706562729 - ...

⌕

run 31b950f4

⌕

Reading

●

HOW IT WORKS

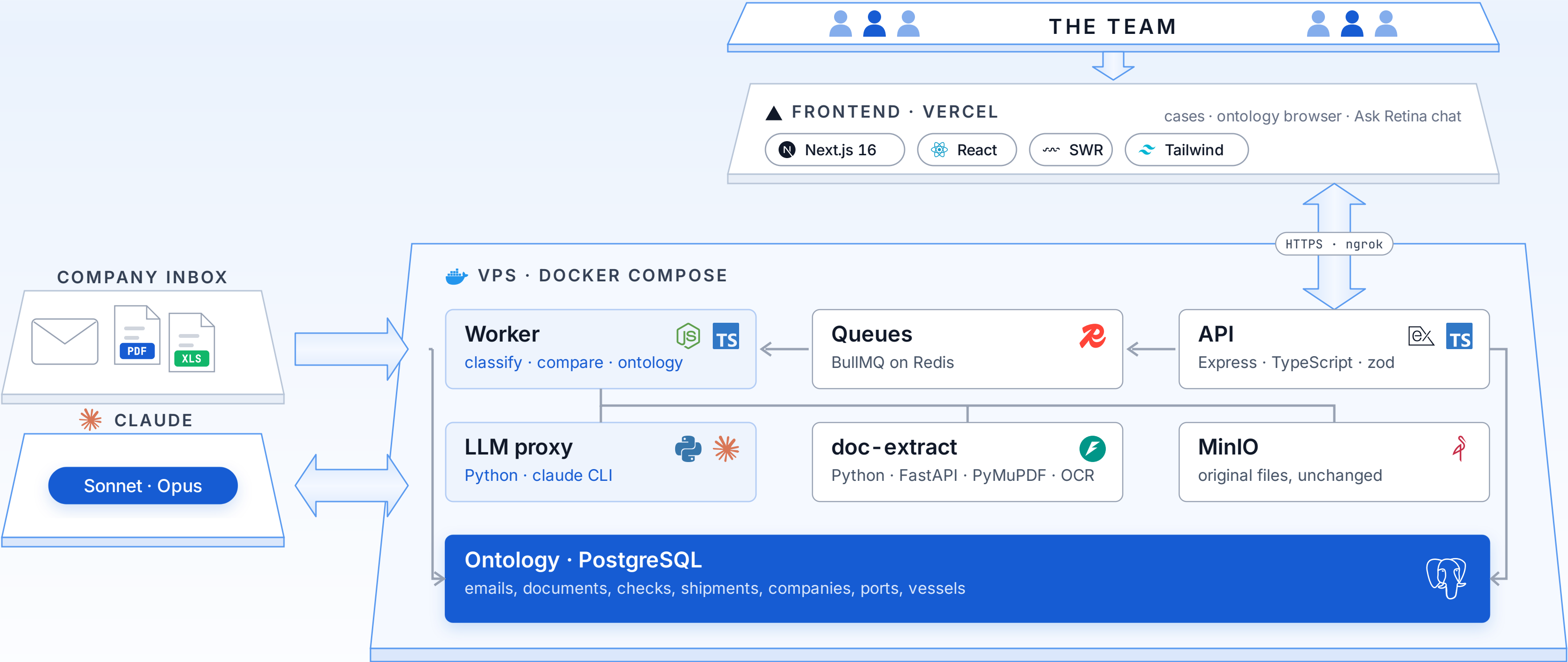
The email pipeline, in three steps



The result of every step is stored, and every value keeps the line it was read from.

ARCHITECTURE

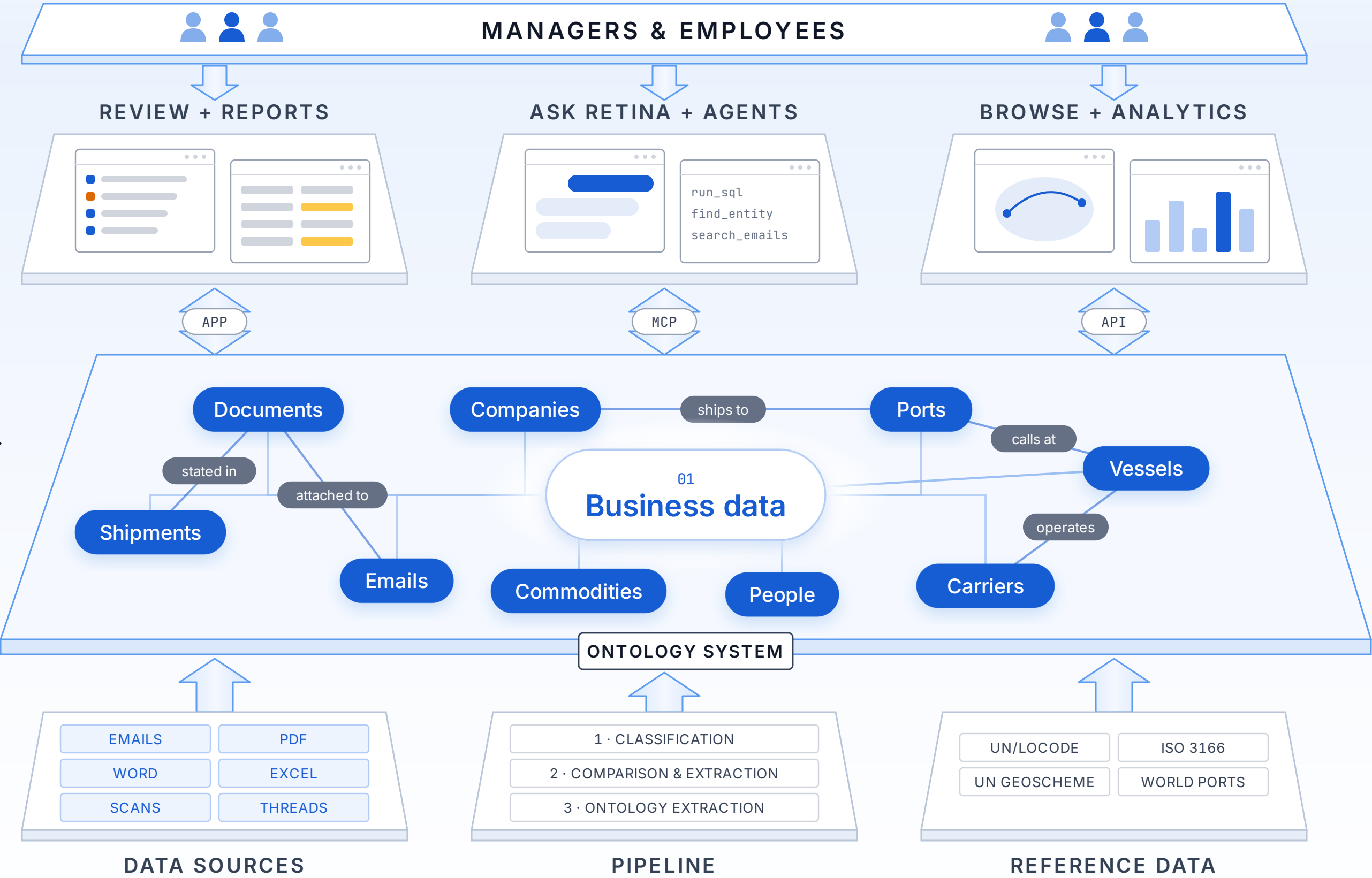
Tech stack and architecture



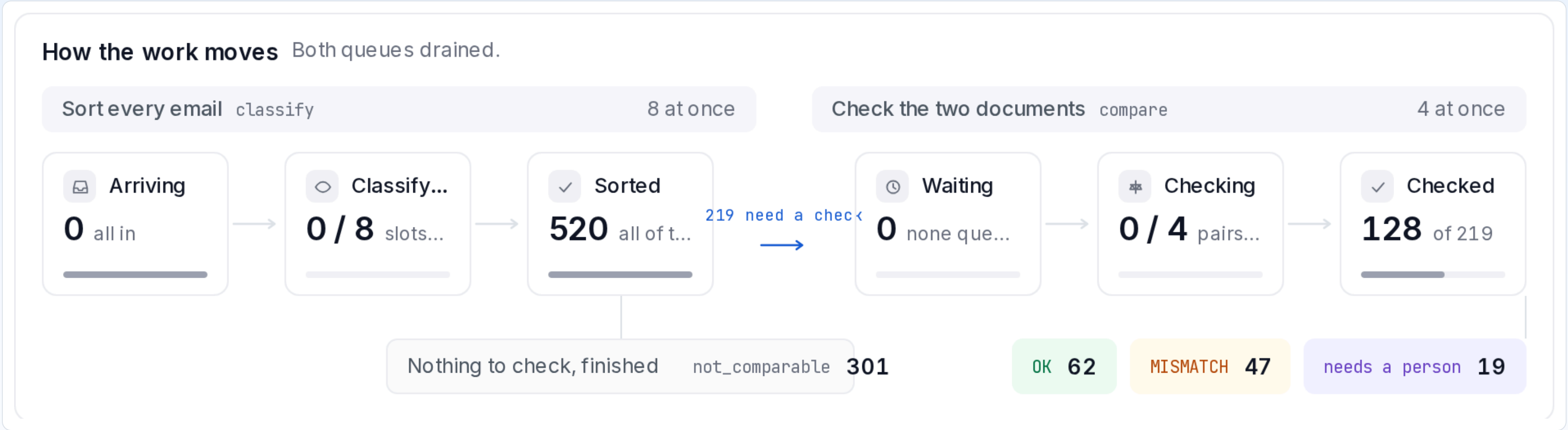
 where a model reads  code and infrastructure Only the API leaves the server. Every external system sits behind one interface.

Where the Ontology Layer sits

Ontology-powered insights & decisions →



A run replays the inbox, and the page shows the work moving



Two queues, watched live

Every email is sorted. Only a comparison request crosses into the second queue to have its documents checked.

Every run is its own experiment

A run pins a prompt version and a model per step, so two prompts are compared on the same 520 emails.

What it took, on the page

520 emails in 30 minutes, 1,601 model calls, 15 percent settled by the verifier, and what it would have cost.

One email, checked: the verdict unfolds back to the quoted line

sales

sales@roxcel.at to ops@aprilasia.com

Dear Arlene, Pls assist to check the draft BL against the SI for PO and revert with any discrepancy asap. Draft BL No. SIN204711671. Best Regards, Teo Ei Leen Shipping Documentation DID : +971 04 4938294 APRIL Fine Paper Trading (Middle East) Fze #813, 4 EA, Dubai Airport Free Zone P.O. Box : 293775, Dubai, United Arab Emirates Website : www.aprilasia.com | www.paperone.com _____ From: Chella Perumal <chella.perumal@psabdp.com> Sent: Friday, January 1, 2026 3:26 AM Subject: RE: 5SUS-86999 Please follow the previous instruction. Thank you. From: Mitchell Ting <mitchelle_ting@aprilasia.com> Sent: Friday, December 24, 2026 2:40 PM Subject: RE: 5SUS-86999 Please follow the previous instruction. Thank you.

email_025_SI.txt

655 B

email_025_BL.txt

649 B

Below this line is Retina, not the sender

Both files opened and were the documents their names claimed. 5 of the 7 fields agree, 2 differ.

sorted BL_COMPARISON

sure 0.98

decided by llm

fields that differ 2 of 7

What differs 2

port_of_discharge

differ

SI

Port of Discharge (POD): FREMANTLE, AUSTRALIA (AUFRE)

BL

Discharge Port: BUSAN, SOUTH KOREA (AUFRE)

SI names FREMANTLE, AUSTRALIA while BL names BUSAN, SOUTH KOREA; a place is defined by its name, and these are different places despite sharing the same (inconsistent) code AUFRE. 0.95 sure

A seam between the sender and Retina

Above the line is the mail as it arrived. Below it is what Retina made of the two documents, in one plain sentence.

The differing words, on both sides

Fremantle in the instruction, Busan in the draft, both under the same stale code AUFRE. The judge names the place, with its confidence.

The other five agree, folded away

A tab shows both documents field by field, and another lists every model call behind this verdict with its prompt version, latency and tokens.

Retina · 09

A difference and an uncertainty are never the same event

RE_AFEMY - VALPARAISO_CHILE - OOCL(OOLU4899718428) - 5RUS...
email_513

NEEDS_REVIEW

The case 1Both documents 2Model calls 8

s

sathiya

sathiya@april.com.my to ops@aprilasia.com

Dear Team, Attached SI and draft BL for PSGSE5969325 for checking (scanned copies (image only)). Please advise.

email_513_SI.pdf

21 KB

email_513_BL.pdf

21 KB

Below this line is Retina, not the sender

2 of the 2 attachments read cleanly. Nothing was guessed from the rest.

parked as unreadable

Retina stopped here

Opened Sep 21, 07:11 PM, nobody assigned

unreadable

Why

email_513_SI.pdf has no text in it, so its 1 page was looked at, and what is on them could not be made out. Nothing a person opening this file could read either, so no value was taken from it.

unreadable

A scanned page came back from OCR under the floor. Retina stops, says why, and asks for a readable copy.

RE_AFEMY - CONAKRY_GUINEA - MONTER(MCLSIN6123859) - 5RCY...
email_516

NEEDS_REVIEW

The case 1Both documents 2Model calls 8

ET

elisa tukiman

elisa_tukiman@april.com.my to ops@aprilasia.com

Dear Team, Please compare the SI and draft BL for MCLSINJEA2551434. Some SI fields were left blank by the customer; kindly confirm what we have.

email_516_SI.txt

458 B

email_516_BL.txt

654 B

Below this line is Retina, not the sender

2 of the 2 attachments read cleanly. Nothing was guessed from the rest.

parked as missing_value

Retina stopped here

Opened Sep 21, 07:11 PM, nobody assigned

missing_value

Why

A field the comparison needs is blank in one of the documents, or a placeholder stands where its value should be. A blank is an uncertainty and never a difference, so the pair was not judged on it.

missing_value

A field left blank by the customer. A blank is not a mismatch, so the pair is not judged on it.

What a person can do

Confirm. Correct a value.
Reclassify. Upload a missing or readable file. Retry a failed job. Add a note.

A correction sits beside the model's reading, never over it, and the pair is judged again from both sides.

Every action is an append-only, labelled record of what a person decided and why.

The mail becomes things, and every value says who wrote it

What Retina knows

Emails520

Ports33

Parties31

Shipments208

Carriers8

Vessels10

Commodities7

People36

Ontology / Emails / email_004

Things

Record

Links

Email

MISMATCH

Open it outside the ontology

email_004

One record of 11 stored values and 9 links. Every value says who wrote it.

Stored values

pk	id	email_004	the source
date	received_at	21 Sep 2026, 11:03	the source
abc	sender	docs@vitalsolutions.sg	the source
abc	subject	REQUEST BL DRAFT _PO 26067_ COATED IVORY BOARD__138MT	the source
enum	category	BL_COMPARISON	a model
1.0	confidence	0.92	a model
enum	decided_by	llm	code
123	attachment_count	2	code
enum	status	MISMATCH	code
list	defect_fields	consignee, notify_party	code
enum	review_reason	not set	nothing wrote it

The record: **the source** wrote the sender, **a model** wrote the category, **code** wrote the status.

Ontology / Emails / email_004

Things

Record

Links

Fit

Two hops

What it holds

id	email_004
received_at	21 Sep 2026, 11:03
sender	docs@vitalsolutions.sg
subject	REQUEST BL DRAFT _PO 26...
category	BL_COMPARISON
confidence	0.92

Step to

one click each

Shipment	1	>
Attachments	2	>
Documents	2	>
Fields	14	>
Differences	2	>
Comparison	1	>
Client	1	>
Same client, differed	0	>
Same consignee, anywhere	18	>

React Flow

email_004

Email, BL_COMPARIS...

carries

judged intc

carries

judged intc

sent it

shipper

rt_of_load

notify_party

vitalsolutions.sg

Client

APRIL FAR EAST (...)

Party, seen 67 times

NANTONG, CHINA...

Port, seen 56 times

EAST BRIGHT FZ-...

Party, seen 34 times

email_004_BL.txt

Document, 7 values read

email_004_S1.txt

Document, 7 values read

The links: who sent it, who ships, which port, which documents, two hops out.

WHAT RETINA KNOWS, THIS RUN

Emails520

Shipments208

Ports33

Parties31

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Vessels10

Carriers8

Nobody typed any of them in.

Companies, ports and shipments get their own pages

Companies / APRIL FINE PAPER TRADING

Company

APRIL FINE PAPER TRADING

Edit

Same as

trading company Malaysia paper trading part of APRIL Group (unverified)

What it is

April Fine Paper Trading is a paper trading company. In this mailbox it appears once, as the shipper on a single email covering a Singapore-to-Yangon shipment of Fujito PaperOne inkjet paper.

What our mail shows

The dossier shows April Fine Paper Trading under one spelling, appearing once, in the role of shipper on 1 email. That email places it on the lane Singapore (SGSIN) to Yangon, Myanmar (MMRGN), carrying Fujito PaperOne inkjet paper aboard the vessel Pacific Sun 1 V.251073E. It is named alongside Asia Pacific Paperboard Trading Pte Ltd, East Bright FZ-LLC, and International Forest Products LLC, and the contact names Mitchell Ting and elisa_tukiman@april.com.my appear on the same email. No physical address has been recorded for it, and the email carries no date in the dossier.

General knowledge, unverified (confidence 0.55)

The name and the associated contact email domain (april.com.my), together with the PaperOne-branded commodity being shipped, point to a link with APRIL Group (Asia Pacific Resources International Limited), a large Indonesia-based pulp and paper producer whose PaperOne brand is well known; "April Fine Paper Trading" would fit as a regional trading/marketing entity for that group, likely based in Malaysia. This is inference from naming and brand association, not a confirmed corporate identity.

This profile is older than the mail that last touched the thing; it is queued for a rewrite.

Roles

shipper 37 emails

Ports it ships through

☐ NHAVA SHEVA, INDIA (INNSA)	9
☐ NANTONG, CHINA (CNNTG)	8
☐ BUATAN, INDONESIA (IDBUA)	7
☐ PORT KLANG (WESTPORT), MALAYSIA (MYPKG)	5
☐ MERSIN, TURKEY (TRMER)	3
☐ SINGAPORE (SGSIN)	3
☐ VALPARAISO, CHILE	3
☐ APAPA, NIGERIA (NGAPP)	2
☐ CALLAO, PERU (PECLL)	2
☐ CONAKRY, GUINEA (GNCKY)	2
☐ FREMANTLE, AUSTRALIA (AUFRE)	2



Port

PORT KLANG (WESTPORT), MALAYSIA (MYPKG)

Edit

Same as

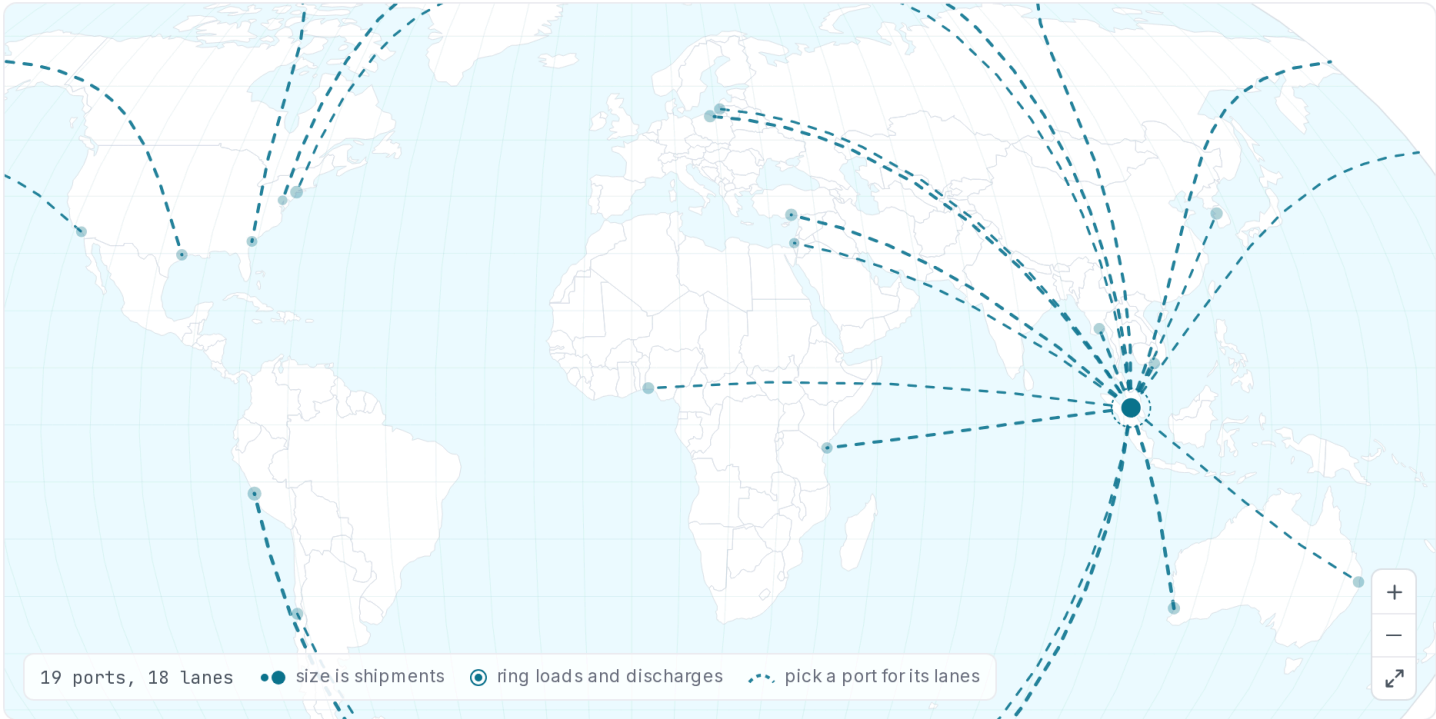
MYPKG

Malaysia

South-eastern Asia

Strait of Malacca

Where it ships 18 Lanes



19 ports, 18 lanes

● size is shipments

⦿ ring loads and discharges

--- pick a port for its lanes

+

-

↻

3, 101.4 · placed by the world's port list

A company: what our mail shows, kept apart from general knowledge marked **unverified**. Roles, lanes, the people on its mail.

A port: its 18 lanes on a map, placed by the world's port list, and the companies that ship through it.

Two spellings of one port are one port, joined only by a judge's verdict or a shared UN/LOCODE, never by string matching. A person can rename, merge or edit any of them, and the edit outlives every rebuild.

Ask Retina answers over the ontology and shows its work

Conversations

New

Ask Retina / what are the shipments to the US? when?

run bcb018bf

what are the shipments to the U...

21 Sep 13:26 · 4 turns

run bcb018bf

What skills do you have? Explain...

21 Sep 12:13 · 4 turns

run bcb018bf

What skills do you have? Explain...

21 Sep 12:10 · 2 turns

run bcb018bf

Give me an overview of this co...

21 Sep 12:06 · 2 turns

run bcb018bf

Give me an overview of this co...

21 Sep 12:04 · 2 turns

run bcb018bf

Give me an overview of this co...

21 Sep 12:00 · 2 turns

run bcb018bf

What is a client information. Wha...

21 Sep 11:55 · 2 turns

run bcb018bf

Which client had the most mism...

21 Sep 11:40 · 2 turns

run bcb018bf

what data do you have in memor...

21 Sep 11:26 · 4 turns

run 62753236

what are the latest shipping for t...

21 Sep 10:31 · 4 turns

every run

A new question

21 Sep 10:28 · 0 turns

run 62753236

Give me the list of this company...

21 Sep 10:26 · 2 turns

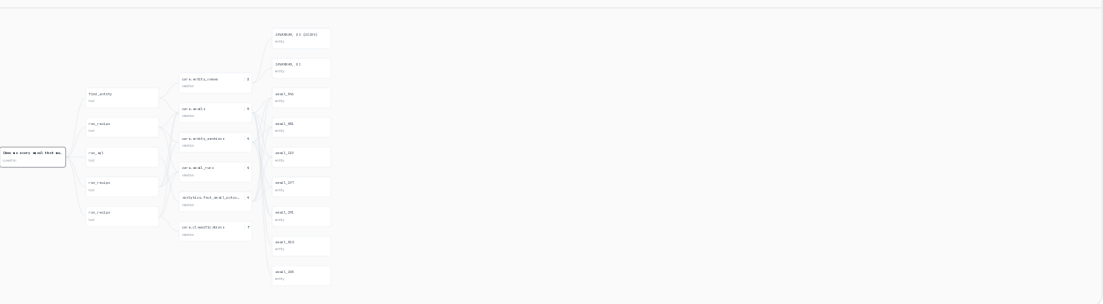
every run

How many ports are there?

21 Sep 10:25 · 2 turns

every run

What it touched 21 nodes, drawn from what the tools reported



Expand

Eight emails in run bcb018bf name Savannah, and all eight are comparison requests.

Savannah as the resolved discharge port

- email_046, roxcel.at, the documents differed, MISMATCH
- email_051, psabdp.com, the documents agreed, OK
- email_119, april.com.my, the documents differed, MISMATCH
- email_197, algurg.ae, the documents agreed, OK . Its subject is about Rugao/Nantong/Shanghai, so this one shows up only in the discharge field.
- email_291, aprilasia.com, under the second spelling SAVANNAH, US without the code

Savannah only in the subject line

- email_018, aprilasia.com, KPP-ANTALIS (SINGAPORE) PTE. LTD.
- email_265, fujitogrp.com, SAFQA LIMITED
- email_299, aprilasia.com, NAGAPPA EXPORTS

The port is spelled two ways here, SAVANNAH, US (USSAV) and SAVANNAH, US, held as two separate things, so I took both. I could not read the comparison outcome for email_018, email_265, email_291 and email_299 on this turn; all four are sorted as comparisons.

Part of it. Looked in: resolved discharge ports, both Savannah spellings, subject lines, sender domains.

Working: 4 queries, 5 calls

Savannah discharge emails 5 Fields that differed Outcomes for the other four

Grounded, then answered

Twelve tools over a read-only role: find a thing, run a recipe, run guarded SQL, explain a decision. The queries and their rows sit under every answer.

Honest about coverage

"Part of it. Looked in: resolved ports, both spellings, subject lines." A total it could not finish is stated as a lower bound.

It follows you

The dock is open on every page and reads what the page is about: this email, this company, this port. "Why did this need a person?" is one click.

Ask Retina

what are the shipments to the US? when?

all eight are comparison requests.

Savannah as the resolved discharge port

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- email_051, psabdp.com, the documents agreed, OK
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Reading run bcb018bf

Ask about what you are looking at

Every call on record, every run scored the way the judges do

RE_ TO CONFIRM DOCS _ 5SUS-86999 _ FREMANTLE_AUSTRALIA _ CERI...
email_025

MISMATCH

The check	7	Both documents	2	Model calls	7
classify	v3	sonnet	3.2s	2 in 167	out
doc-type	v1	sonnet	5.8s	2 in 355	out
doc-type	v1	sonnet	6.6s	4 in 500	out
extract	v1	sonnet	13.9s	4 in 1681	out
extract	v1	sonnet	14.7s	4 in 1742	out
field-judge	v1	sonnet	12.4s	4 in 1365	out
shipment-read	v1	sonnet	97.2s	2 in 10406	out

Seven calls behind one email: step, prompt version, model, latency, tokens. Every attempt is a row, whether or not it parsed.

A gate in front of the pipeline prices every email from counts, sizes and sender history before a model sees it, and holds what a budget cannot pay for. It never reads the words.

Final score

over 520 emails

0.9975

stage 1

macro F1 over the five categories

0.9987

0.30

stage 3

defect F1 over the comparable pairs

0.9892

0.20

end to end

every check right on one email

1.0000

0.50

Stage 1

rows are the truth, columns the answer

	BL_COMPARISON	SI_REQUEST	INVOICE_QUERY	GENERAL	SPAM
BL_COMPARISON	219	1	0	0	0
SI_REQUEST	0	125	0	0	0
INVOICE_QUERY	0	0	75	0	0
GENERAL	0	0	0	60	0
SPAM	0	0	0	0	40

Accuracy 0.9981. Anything off the diagonal is an email sorted into the wrong category.

Escalation

0.9500

wrong_doc_type

4 of 5

missing_attachment

5 of 5

unreadable

5 of 5

missing_value

5 of 5

19 escalated of 20 due, at 1.0000 precision. Stage 3 saw 200 comparable pairs.

The organisers' formula, ported line for line and proven equal to their scorer. A train and holdout split, read last.

Priority by client tier, with tonnage as the tiebreaker and aging so a small client is never starved. A tier decides no category and reaches no prompt.

Four things a script, a rule set, or a plain chatbot cannot do

It reads. It has no rules to break.

No sender lists, no keyword tables, no regex over a body, no label map, no string matching between spellings. A fresh inbox with a different seed works the same, because nothing was fitted to this one.

Every verdict walks back to the line.

A value without a quote in the document is dropped, not stored. A join between spellings needs a verdict. A profile separates what the mail shows from what a model believes. Trust on a document of title needs exactly this.

The check is the entry. The ontology is the product.

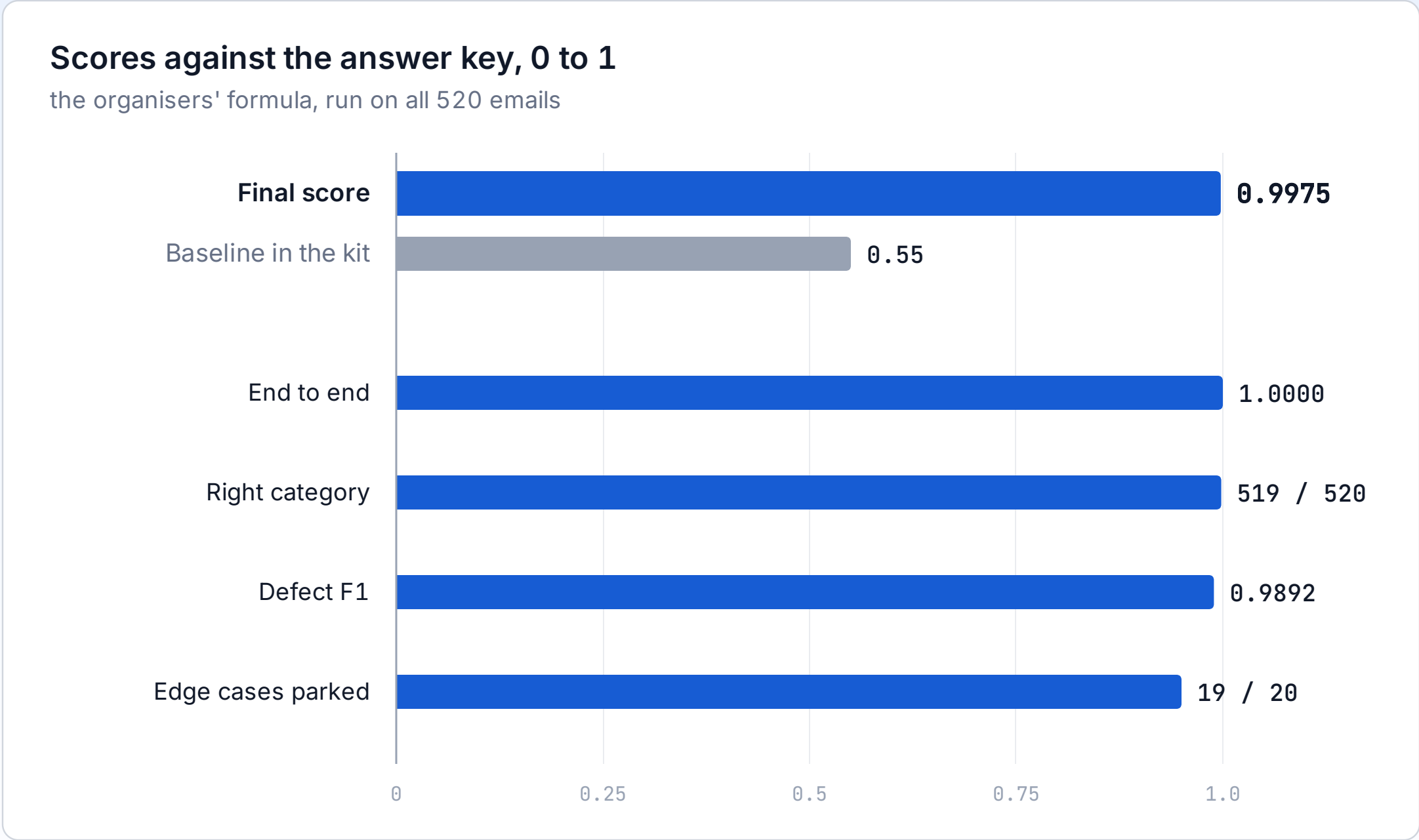
Checking documents produces typed companies, ports, lanes, vessels and shipments as a side effect. The same reading answers "who ships to Savannah" without another integration.

It is measured, priced and gated.

A prompt change is a run with a score. A call is a row with a cost. An email is priced before a model reads it. A person can see every decision and reverse it. That is what makes it deployable, not a demo.

RESULTS

The full 520-email inbox, scored with the organisers' formula



0.9975

final score, against a 0.55 baseline shipped with the kit

30 min · 1,601 calls

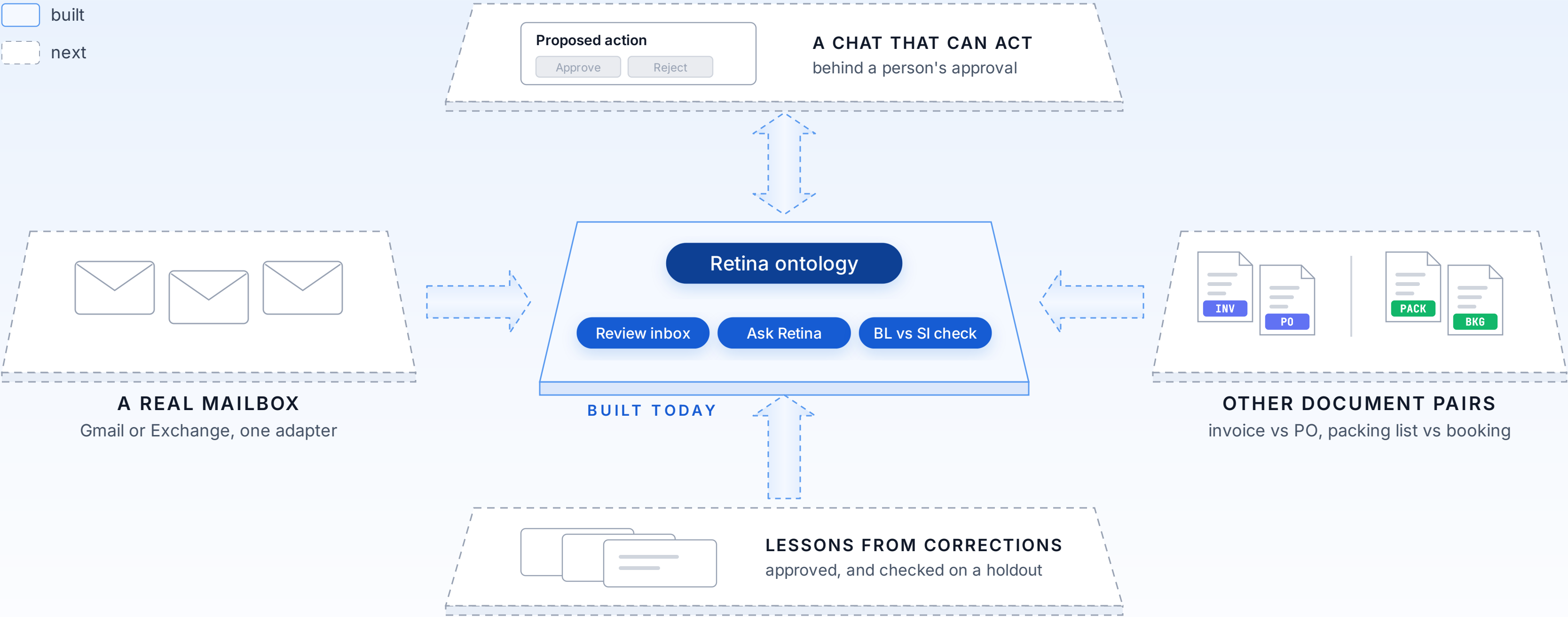
for 520 emails, 8 and 4 in parallel, \$57 at list price

0 rules

fitted to the dataset, so a fresh seed is the same job

WHAT IS NEXT

Built for one inbox. Designed for any



The ontology, the chat and the review inbox stay the same for any company communication. Only the document pair and the enums are shipping.

Your mail already holds your business data. Retina makes it readable, checkable and askable.

**Readable**

every email and attachment

**Checkable**

every field, back to its line

**Askable**

by anyone, in plain words



